

DEEP DIVE • CASE 018

The temporary lifting detail was the gravity system

L'Ambiance Plaza • Bridgeport, Connecticut • 1987 • Lift-slab construction

TEMPORARY WORKS

LOAD PATH

SUPPORT LOSS

CONSTRUCTION STAGE

On 23 April 1987, the partially completed L'Ambiance Plaza apartment complex collapsed during lift-slab construction. The National Bureau of Standards investigation reconstructed the load path at the instant of failure and tested the hardware that was then holding the slabs in the air.

Its conclusion was specific: the most probable initiating event was loss of support at one lifting jack in the west tower while an upper package of three floor slabs was being positioned. The investigation reproduced a mechanism in which a lifting angle in the shearhead assembly deformed excessively and the lifting nut slipped free.

YOU ARE THE ENGINEER

A contractor proposes lifting a multi-level slab package using a specialist system. The permanent structure has been checked thoroughly. What do you ask for before the lift?

A weak answer is **the jack capacity**. You need the temporary structural model: where load enters each lifting point; how tolerances and deformation affect bearing; how the package is stabilised; what happens if one point moves differently; whether redistribution is safe; what bracing exists; and which observations trigger a stop.

THE CONSTRUCTION STAGE IS A SEPARATE STRUCTURE

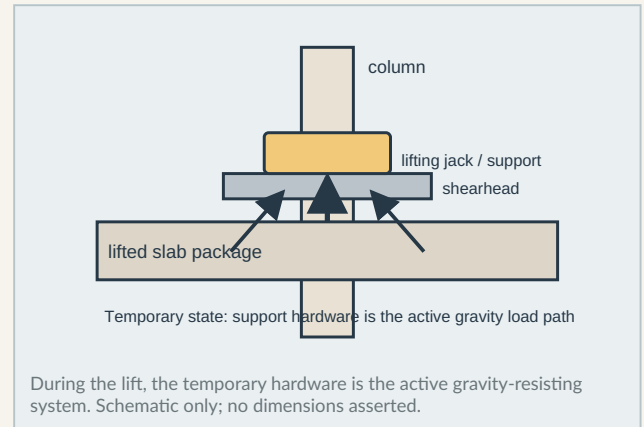
Construction-stage design is often treated as a reduced version of the final building. That can be dangerously wrong. The topology itself may differ: supports exist that vanish later; permanent diaphragms may not yet be engaged; dead load can be concentrated in lifted packages; temporary attachments introduce eccentricity and local deformation modes never present in service.

So the correct question is not **is the final building strong enough?** It is **what structure exists right now?**

Formwork, falsework, launching equipment and erection bracing deserve the same logic. Draw the actual load path for each critical stage. If a stage depends on a few temporary points, test the consequence of one point losing stiffness, bearing or alignment. If redistribution is relied upon, show where the force goes.

ENGINEER'S NOTEBOOK • NOTE-005

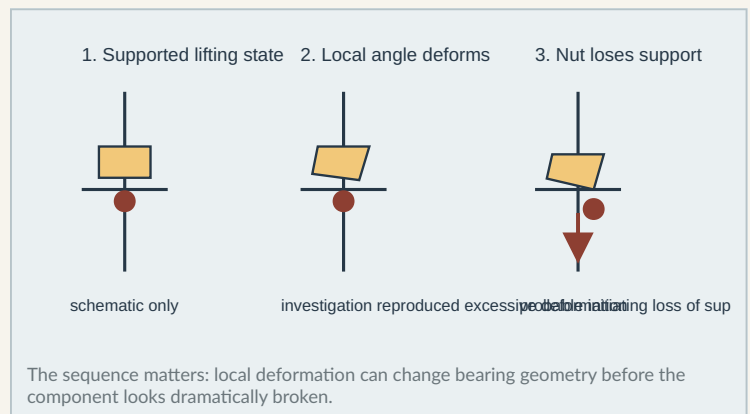
Temporary works are the structure until the permanent path exists. Review lifting, propping and transfer states as independent structural systems. Check deformation compatibility, support loss, stability, redistribution and stop-work criteria.



RECONSTRUCT THE STATE, NOT THE FINISHED DRAWING

The slab package delivers gravity load into the lifting points. Those forces pass through the jack/support hardware and shearhead into the columns. Safety depends on each active lifting point retaining a reliable force path while the slabs are moving.

Lift-slab construction reverses the usual mental hierarchy. In the completed building, slabs are supported permanently by columns and their connections. During lifting, the temporary arrangement is not merely means and methods beside the structure. **It is the structure carrying gravity at that moment.**



The investigation mechanism can be read as three states: supported lifting, excessive local deformation of the lifting angle, and loss of nut support. A component can remain recognisable and yet cease to perform the geometric job that lets another component bear. Once the support condition changes, the force path changes immediately.

EVIDENCE BOUNDARY

The diagrams omit dimensions, jack loads and reinforcement quantities not directly extracted from the investigation. The mechanism is documented; invented precision would make the explanation look stronger while making it less true.

SOURCES FOR THIS CASE

Primary • Tier A — NBS/NIST, *Investigation of L'Ambiance Plaza Building Collapse in Bridgeport, Connecticut* (NBS IR 87-3640), 1987. **NIST publication**

SITE PROBLEM · CASE 019

019

Water beneath the slab was a structural load

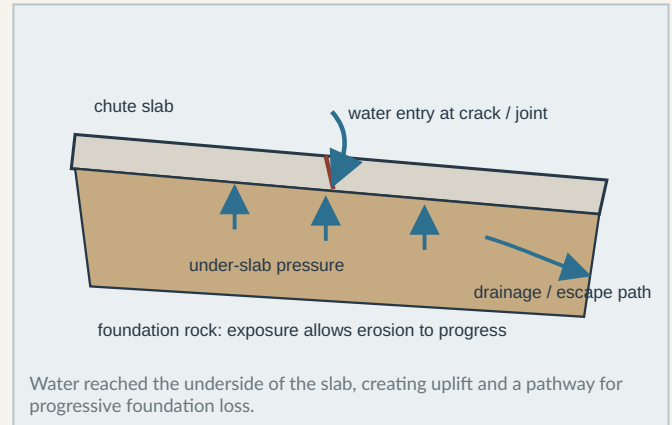
Oroville Dam Spillway · California · 2017 incident / 2018 forensic report

The Oroville service spillway failure is enormous in scale, but its transferable lesson fits on an ordinary site sketch. The independent forensic investigation described water entering beneath the concrete chute slab through cracks and joints. Where relief was inadequate, that water could create uplift pressures beyond the slab system's available resistance. Once local slab damage exposed the foundation, high-velocity flow attacked variable-quality rock and erosion progressed rapidly.

YOU ARE THE ENGINEER

A sloping concrete apron has recurring cracks and a history of wet ground. The surface looks serviceable. Do you simply seal the cracks? **Not until the pressure path is understood.**

Sealing the top may reduce entry, but it can hide symptoms while water continues to arrive from groundwater or upslope sources. Establish where the water comes from, whether a continuous relief path exists, whether outlets are maintainable, and whether voiding has already developed.



Water reached the underside of the slab, creating uplift and a pathway for progressive foundation loss.

The mechanism is easy to miss because engineers often draw water above concrete and loads below it. Here, water reached the underside. The slab then had hydraulic pressure trying to lift it while flow through defects and damaged areas created a pathway for progressive foundation loss.

ENGINEER'S NOTEBOOK

Ask how water gets in, then ask how it gets out. The second question is often the structural one.

For an ordinary slab, apron, channel or retaining structure, the same questions survive: Where can water enter? Can it drain faster than it enters? What pressure develops if relief is blocked?

EVIDENCE BOUNDARY

Oroville-specific uplift values, drain geometry and rock properties are not generic design inputs. This transfers a mechanism and a diagnostic habit, not a ready-made pressure coefficient.

SOURCES FOR THIS CASE

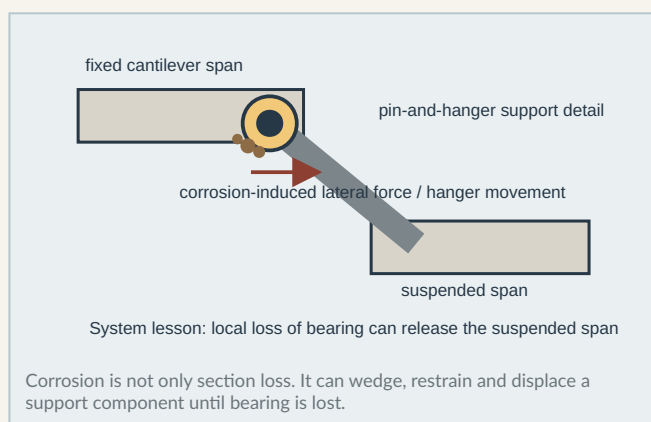
Primary · Tier A — Independent Forensic Team / California DWR, Oroville Dam Spillway Incident — Final Report, 5 Jan 2018. **Forensic Team**

DETAIL / SUPPORT · CASE 020

020

A compact support detail lost the span

Mianus River Bridge · Connecticut · 1983 · Pin-and-hanger suspended span



DETAIL CHECKLIST

Trace four things together: **force, movement, water and inspection access.** A drawing that shows only the first two is unfinished.

The Mianus River Bridge collapse is a reminder that a structure can be vast while its governing failure surface is small. The NTSB concluded that a suspended highway span lost support after lateral displacement of a hanger in a pin-and-hanger assembly. Corrosion-induced forces contributed to that movement, and inspection and maintenance deficiencies allowed the deteriorating condition to remain undetected.

A pin-and-hanger detail works elegantly on a clean drawing, but it must survive decades of water, debris, corrosion products, movement and access constraints. Corrosion does more than reduce net section. Expanding products can push, wedge and restrain components; intended movement can be impaired while unintended lateral forces grow.

Ask whether the detail remains **inspectable in its governing failure mode**. Can a person see bearing surfaces? Can debris be removed? Can displacement be measured? Can water drain? Is there a secondary support if the local component disengages?

Practical transfer: identify the component whose failure would make the rest of the calculation irrelevant, then review it in its dirty, misaligned, partly blocked and difficult-to-inspect real condition. If one small assembly uniquely supports the system, it deserves major-member scrutiny.

SOURCES FOR THIS CASE

Primary · Tier A — NTSB, Collapse of a Suspended Span of Interstate Route 95 Highway Bridge over the Mianus River, DCA83AH003. **NTSB investigation**

STRUCTURAL ENGINEERING WIN · CASE 021

021

Measured shaking, limited damage

Olive View Medical Center · Los Angeles · 1994 Northridge earthquake

Olive View is useful because success was instrumented. The replacement hospital had been designed in 1976 with deliberately high lateral strength and stiffness. During the Northridge earthquake, USGS instruments recorded about **0.91 g** in the free field, **0.82 g** at the ground floor and **2.31 g** at the roof. Despite that severe recorded response, the building experienced limited structural and nonstructural damage.

0.91 g

free-field record

0.82 g

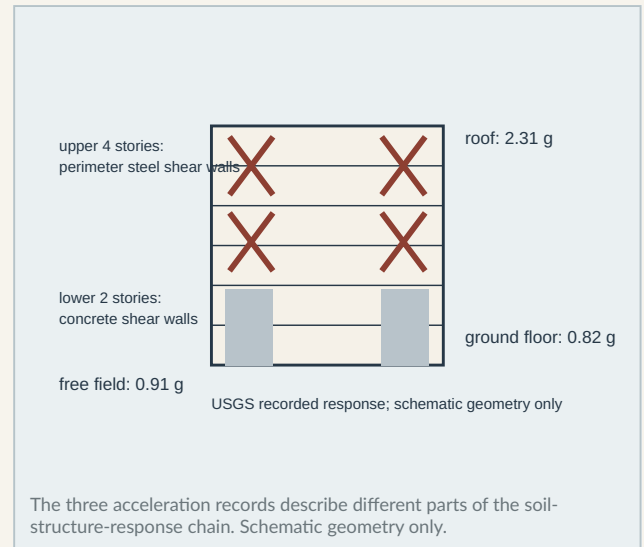
ground-floor record

2.31 g

roof record

The lateral system was not exotic. The USGS paper describes concrete shear walls in the lower two stories and perimeter steel shear walls in the upper four. What makes the case valuable is the evidence chain: **known structural system** → **recorded ground motion** → **recorded building response** → **observed performance**.

That is much more useful than saying a building “met code.” A code check predicts performance under modelling assumptions. Instrumentation gives the structure a chance to answer back.



The three acceleration records describe different parts of the soil-structure-response chain. Schematic geometry only.

THREE NUMBERS, THREE DIFFERENT QUESTIONS

The free-field record describes shaking arriving at the site. The ground-floor record shows how the base of the building participated in that motion. The roof record shows the dynamic response of the structure itself.

The roof acceleration being larger than the ground motion is not a contradiction. Structures filter and amplify motion according to stiffness, mass, damping and modal response. The value of the case is not the isolated number 2.31 g. It is the ability to connect measured demand to an actual lateral system and then to observed damage.

MEASUREMENT IS PART OF ENGINEERING

For ordinary projects, full strong-motion instrumentation may be unrealistic. The principle still transfers. Build feedback into systems where consequence warrants it: movement gauges at retaining walls, crack monitors, settlement points, inclinometers, load cells on temporary works, piezometers where pore pressure matters, or survey targets where a slope or excavation is expected to move.

A MONITORING PLAN NEEDS AN ACTION

A useful monitoring plan answers four questions before installation: **What quantity matters? What change is meaningful? Who receives the reading? What action follows a trigger?**

A sensor without an action threshold is merely a very diligent historian. Monitoring changes how uncertainty is managed: instead of pretending the model has removed uncertainty, the design can acknowledge a range of plausible behaviour and use observations to determine which branch reality has chosen.

That is particularly valuable in staged construction, ground movement and temporary works, where the system evolves with time and the assumptions that mattered at design may be testable during construction.

ENGINEER'S NOTEBOOK

Prefer measured performance whenever important assumptions can be instrumented. The objective is not data collection for its own sake; it is to close the loop between predicted behaviour, observed behaviour and the decision that follows.

SOURCES FOR THIS CASE

Primary · Tier A — U.S. Geological Survey, M. Çelebi, *Response of Olive View Hospital to Northridge and Whittier earthquakes*, 1997. [USGS publication](#)

RETAINING WALL

Survey targets, movement gauge or crack monitor where displacement matters.

TEMPORARY WORKS

Load, level or movement measurement tied to a stop-work threshold.

SLOPE / EXCAVATION

Survey or inclinometer data where progressive ground movement is plausible.

GROUNDWATER

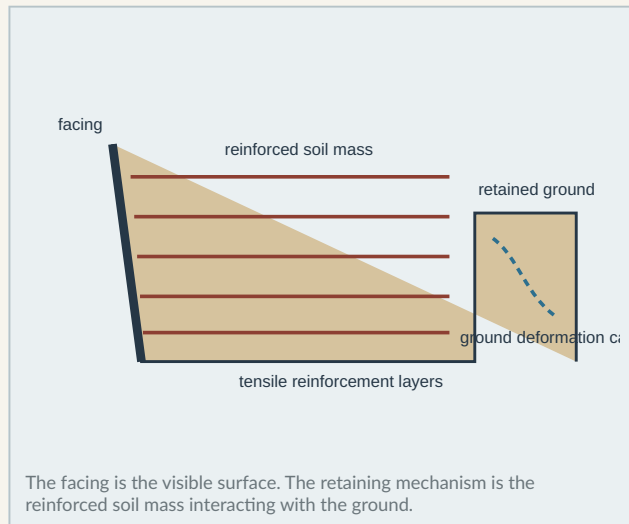
Piezometric observation where pore pressure is part of the governing mechanism.

GEOTECHNICAL ENGINEERING WIN · CASE 022

022

The reinforced soil mass behaved as a system

Maule earthquake · Chile · 27 February 2010



ENGINEER'S NOTEBOOK

Treat the reinforced soil zone as the structure.

Review reinforcement continuity and length, fill quality, compaction, drainage, foundation behaviour and global stability as one system.

FHWA's post-earthquake reconnaissance reported that all mechanically stabilized earth walls observed by the team exceeded expected performance. That statement is especially useful because the same reconnaissance documented settlement, liquefaction and lateral-spreading effects elsewhere in the transportation network.

An MSE wall works by creating a composite mass. Layers of tensile reinforcement interact with compacted fill so the reinforced zone behaves as a coherent block, while the facing provides local retention and serviceability. The system does not ask a thin wall stem to resist all lateral demand through bending at one base section.

That distributed mechanism can be forgiving. Reinforcement engages over many layers and a large soil volume, while the system can accommodate some deformation without requiring the entire retained mass to behave as a rigid body. In seismic loading, that distinction matters because an earth-retaining system is asked to survive both inertial demand and whatever deformation the foundation and surrounding ground impose.

FACING

Provides local retention and serviceability. It is the visible surface, not the whole gravity-resisting mechanism.

REINFORCED FILL

Tensile reinforcement and compacted soil act together to form the reinforced block.

STILL EXTERNAL

Foundation behaviour, global stability, drainage and deformation outside the reinforced zone remain governing checks.

THE WIN IS EVIDENCE, NOT A MYTH

"MSE walls did well in Maule" does not mean every reinforced-soil wall is seismically invulnerable. Foundation failure, global stability, poor fill, inadequate reinforcement length, drainage problems or construction defects remain available.

The transferable lesson is that **distributed reinforcement can produce a resilient soil-structure system**, and real earthquake reconnaissance gives that mechanism empirical weight.

COMPARE SYSTEMS BY CONSEQUENCE

For retaining-wall design, compare systems by failure consequence as well as calculated factor of safety. Ask which arrangement can accommodate deformation, which concentrates demand, what happens if drainage degrades, whether the foundation can move, and whether construction quality can be verified layer by layer.

SOURCES FOR THIS CASE

Primary · Tier A — FHWA, *Post-Earthquake Reconnaissance Report on Transportation Infrastructure: Impact of the February 27, 2010, Offshore Maule Earthquake in Chile*, FHWA-HRT-11-030, 2011. [FHWA report](#)

THE THREAD · NOTHING IS SECONDARY

L'Ambiance says the temporary support is the structure while the slab is airborne. Oroville says drainage becomes structural when trapped water can lift the slab. Mianus says the small support detail governs when no alternate path exists.

Olive View says the lateral system earns trust through measured performance. Maule says soil reinforcement matters because it changes the behaviour of the whole retained mass.

60-SECOND TAKEAWAY

- Draw the load path for the construction state, not only the completed structure.
- Treat trapped water as a potential load and drainage as pressure control.
- Detail critical supports for inspection, movement and deterioration.
- Prefer measured performance where assumptions can be instrumented.
- Distributed soil reinforcement can change the governing mechanism more than simply making a wall thicker.

ARCHIVE RECALL

Which earlier Casebook issue showed another case where the apparent main member was not the governing problem, and what interface actually controlled the failure?

Common habit: rank components by consequence, not by visual prominence or contractual label.